

1. Administrative & Study Identification

Full Study Title: Phase 3 Study to Evaluate Two Regimens of Ianalumab on Top of Standard-of-care Therapy in Patients With Systemic Lupus Erythematosus (SIRIUS-SLE 1) **Short Title/Acronym:** SIRIUS-SLE 1 **Protocol Number:** CVAY736F12301 **NCT Number:** NCT05639114 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Ianalumab (VAY736) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of ianalumab (administered monthly or quarterly) compared to placebo in achieving an SLE Responder Index-4 (SRI-4) response with sustained steroid reduction on top of standard-of-care treatment in participants with active systemic lupus erythematosus (SLE). **Secondary Objectives:** To evaluate BICLA response, achievement of Lupus Low Disease Activity State (LLDAS), safety, tolerability, and reduction in total B cell counts and B-cell related genes.

2.2 Background & Rationale To address the unmet need in systemic lupus erythematosus for patients who continue to experience active disease despite standard-of-care therapy (e.g., corticosteroids, anti-malarials, or DMARDs). Ianalumab is a novel biologic that acts via a dual mechanism—blocking the B-cell activating factor (BAFF) receptor and directly depleting B cells—potentially offering a more targeted and comprehensive immune system reset for SLE patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (Active treatment on top of SoC vs. Placebo on top of SoC) **Blinding:** Double-blind **Randomization:** Randomized, 3-Arm Parallel Group (monthly ianalumab, quarterly ianalumab, monthly placebo)

3.2 Planned Enrollment & Duration Target \$N\$: 406 participants **Number of Sites:** Multicenter (United States + 18 other countries) **Estimated Study Start:** 01-Mar-2023 **Estimated Primary Completion:** 19-Apr-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants aged 12 to 100 years. 2. Diagnosis of systemic lupus erythematosus meeting the 2019 EULAR/ACR classification criteria at least 6 months prior to screening. 3. Elevated serum titers at screening of anti-nuclear antibodies (ANA) $\geq 1:80$ and currently receiving standard-of-care treatment (corticosteroids and/or anti-malarial treatment and/or another DMARD).

4.2 Exclusion Criteria 1. Pregnant or nursing (lactating) women, and women of child-bearing potential (WOCBP) not using highly effective contraception. 2. Presence of severe lupus kidney disease (proteinuria > 2 g/day or serum creatinine > 2.0 mg/dL). 3. Chronic infection with HBV, HCV, or active tuberculosis; or abnormal labs at screening (platelets $< 25,000/\text{mm}^3$, Hgb < 8.0 g/dL, or ANC $< 0.8 \times 10^3/\mu\text{L}$).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered either monthly (every 4 weeks) or quarterly (every 12 weeks). Participants in the 12-week group also receive placebo injections on the 4th and 8th weeks of each cycle to maintain blinding.

Route of Administration: Subcutaneous (SC) injection. **Duration of Treatment:** 56 weeks of active dosing for a 60-week primary endpoint evaluation.

5.2 Concomitant & Prohibited Therapy Permitted: Current standard-of-care (SoC) medications for SLE, including stable doses of oral corticosteroids (≤ 30 mg/day), anti-malarials, or protocol-specified DMARDs. **Prohibited:** Cyclophosphamide or biologics within 24 weeks prior to screening; rituximab or other B cell-depleting therapies within 36 weeks; traditional Chinese medicines within 30 days prior to randomization.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, ANA testing, Lab screens
Baseline	Day 0	Randomization, First SC Dose, SoC verification	Interim
Every 4 Weeks	Efficacy Assessments (SLEDAI-2K, BILAG-2004), SC Injections, Safety Labs	End of Study	Week 60 Final Endpoint Evaluation (SRI-4), Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of patients achieving an SLE Responder Index-4 (SRI-4) response with sustained steroid

reduction at Week 60. **Measurement Method:** Composite clinical assessment requiring: a reduction from baseline of ≥ 4 points in SLEDAI-2K, no new BILAG-2004 A or ≥ 2 new B items, and no worsening in Physician Global Assessment of Disease Activity (PhGA) ≥ 0.3 points.

7.2 Secondary & Exploratory Endpoints 1. Proportion of patients achieving BICLA response (Reduction of all baseline BILAG A to B/C/D and baseline B to C/D). 2. Achievement of Lupus Low Disease Activity State (LLDAS). 3. Reduction in total B cells, including naïve and memory B cells, and antibody-producing subsets.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via patient interviews, injection site reaction monitoring, and central laboratory testing. **Serious Adverse Events (SAEs):** Reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External, independent Data Monitoring Committee evaluating interim safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes. Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Target $N=406$ is powered ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to detect statistically significant differences in SRI-4 response rates between either active ianalumab arm and the placebo arm. **Handling of Missing Data:** Standard imputation methods (e.g., Non-Responder Imputation [NRI] for binary composite endpoints like SRI-4 and BICLA).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of trial sites to ensure protocol adherence and accurate AE reporting. **Audit Trail:** eCRF locking, source data verification for primary clinical outcome measures (SLEDAI-2K, BILAG-2004), and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: {{CVAY736F12301}} **Registry Number:** {{NCT05639114}}
Sponsor/Lead Organization: {{Novartis Pharmaceuticals}} **Study Title:** {{SIRIUS-SLE 1}} **Official Regulatory Title:** {{A Randomized, Double-blind, Parallel Group, Placebo-controlled

Multicenter Phase 3 Study to Evaluate Efficacy, Safety and Tolerability of Two Regimens of Ianalumab on Top of Standard-of-care Therapy in Patients With Systemic Lupus Erythematosus (SIRIUS-SLE 1))}

12. Clinical Framework (Systemic Lupus Erythematosus Specific)

Primary Condition: {{Systemic Lupus Erythematosus (SLE)}}
Diagnosis Threshold: {{2019 EULAR/ACR classification criteria + ANA \$\\ge 1:80\$}}
Medical Methodology: {{Standard-of-care Therapy (SoC)}}
Optimization Target: {{BAFF receptor inhibition and B-cell depletion}}

13. Study Procedures & Methodology

Management Pathway: {{Ianalumab SC injection administered alongside SoC}}
Education Protocol (HCP): {{Investigator training on composite endpoint scoring (SLEDAI-2K, BILAG-2004, PhGA)}}
Education Protocol (Patient): {{Adherence to background SoC medications and injection scheduling}}
Quality Audit Mechanism: {{Clinical monitoring and central laboratory tracking for safety and biomarker analysis}}

14. Observation & Follow-Up Schedule

Total Duration: {{60 weeks for primary endpoint evaluation}}
Onsite Visit Cadence: {{Every 4 weeks (Q4W) for injections and clinical assessments}}
Remote Monitoring Frequency: {{As per individual site/protocol requirements}}
Checklist Review Interval: {{Every 12 weeks for major systemic clinical reviews}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				